

AUDITORY GROUPING DAILY

Thursday, 09 July 2026 · 369 papers screened · curated

Today's batch (369 candidates, after folding in four papers found by chasing watch-authors — Auksztulewicz, Németh, Urai/Churchland, and the Neuropixels-Opto team — beyond the automated fetch) has a real throughline: prediction, uncertainty, and boundary detection in auditory and prefrontal circuits, from single cortical layers up to whole-brain fMRI. The Paper of the Day, Zacks and colleagues' dissociation of error-driven versus uncertainty-driven event boundaries in the human brain, is about as direct a formalization of chunking as this pipeline is likely to surface, and pairs naturally with today's laminar-fMRI predictive-auditory-processing paper and the MEG statistical-learning paper. It's also a strong methods day: KIASORT (a drift-robust spike sorter that beats Kilosort4) and Neuropixels Opto (integrated recording plus spatially-addressable optogenetics) are both concrete upgrade paths for the mouse Neuropixels pipeline. One data-quality flag: three eLife candidates (ids 261, 262, 264) carried corrupted DOI fields from the fetcher — verified independently against PubMed, titles/authors/abstracts are correct, only the DOI field itself was wrong.

CHUNKING & SEQUENCE PROCESSING

Environment geometry alters sequential route learning and its integration into cognitive maps.

PUBMED Jaida Long et al. — *Nature Communications* — 2026

Corresponding: Thackery Brown — Georgia Institute of Technology — USA | [Open paper](#)

This study shows that serial-order effects in route learning combine additively with environmental-geometry effects on spatial memory, with the worst location memory occurring at the end of routes through the most irregular (trapezoid) environments. That additive combination of a position-in-sequence effect and an environment-shape effect is a useful precedent for the location-sequence version of the mouse task you're currently developing, where sequence elements are defined by spatial position rather than pitch: it suggests position-dependent memory effects (like the primacy bias you're probing in Protocol 8) and physical/geometric context could combine rather than trade off, which becomes testable once that task has enough conditions to separate the two factors.

AUDITORY & PREFRONTAL ELECTROPHYSIOLOGY

Neuronal dynamics, timing, and flow of sensory and choice-related information in auditory-prefrontal circuitry

RSS:NATURE COMMUNICATIONS *Nature Communications* — 2026-07-08

Corresponding: Bruno B. Averbeck — National Institute of Mental Health (NIH) — United States | [Open paper](#)

Recording from macaque auditory cortex and prefrontal cortex during an auditory decision task, this group shows that raw sensory information flows in the feedforward direction (auditory cortex to prefrontal cortex) while choice-related information flows bidirectionally between the two regions — a clean dissociation of what propagates forward versus what involves both areas jointly. This is close to a direct analogue of the mouse sequence-discrimination task's core question of how auditory cortex and prelimbic cortex jointly support the go/no-go decision: the directional/timing analysis here is a template for asking whether your own Neuropixels recordings from prelimbic and auditory cortex show the same feedforward-sensory/bidirectional-choice pattern around the decision window at the last chord.

Medial prefrontal cortex encodes but is not required to generate goal-directed actions under threat.

PUBMED Muhammad S Sajid et al. — eLife — 2026

Corresponding: Manuel A Castro-Alamancos — University of Connecticut — United States | [Open paper](#)

Using fiber photometry, single-cell calcium imaging, and optogenetic inhibition in mouse medial prefrontal cortex during learned avoidance tasks, this study finds that GABAergic medial-prefrontal neurons encode cues, actions, and outcomes with increasing task complexity, but silencing that cortex has minimal effect on learning or performance — its activity reflects movement and evaluative signals rather than causally generating the action. That's a direct methodological caution for the prelimbic-cortex recordings in the mouse sequence-discrimination task: a region can carry rich task-related information, which your prelimbic Neuropixels data will likely also show, without being causally necessary for the behavior, so any causal claim about prelimbic cortex driving the go/no-go decision would need its own inactivation experiment, not just encoding strength.

Predictive acoustical processing in human cortical layers

CROSSREF (MODEL-DISCOVERED) Lonike K. Faes et al. — Nature Communications — 2026-05-26

Corresponding: Federico De Martino — Maastricht University — Netherlands | [Open paper](#)

Using laminar-resolution fMRI during a cascading auditory oddball paradigm, this study (co-authored by watch-listed predictive-processing researcher Ryszard Aukstulewicz) localizes prediction-error signals to superficial cortical layers of planum polare and middle layers of lateral temporal cortex, with model-updating tied to deep-layer activity — evidence for a layer-specific implementation of hierarchical predictive coding in human auditory cortex. This gives a concrete laminar target for the statistical-learning side of the human chunk-in-stream task: if high-gamma responses during the statistical-background-variation stage of that task (where bottom-up pattern extraction and top-down template matching are meant to dissociate) show a similar superficial-versus-deep split, that would support reading prediction-error and model-update signals off different cortical depths rather than treating high-gamma as one undifferentiated regularity signal.

BEHAVIORAL & COMPUTATIONAL MODELING

Age-related changes in behavioural and neural variability in a decision-making task

CROSSREF (MODEL-DISCOVERED) Fenying Zang et al. — Nature Communications — 2026-06-30

Corresponding: Anne E Urai — Leiden University — Netherlands | [Open paper](#)

Recording over 18,000 neurons across 16 brain regions in young and aged mice performing a visual decision task, this group shows that aging increases post-stimulus neural variability (Fano factor) and reduces stimulus-induced variability quenching, with region-specific direction — higher firing in cortex and striatum, lower in thalamus. This is a large-scale demonstration of exactly the kind of continuous, state-dependent framing your GLM Module C (slow drift from satiety or time-on-task) is trying to capture in the mouse sequence-discrimination task: variability-quenching metrics like theirs could be applied session-by-session to your prelimbic and auditory cortex recordings to ask whether within-session drift in engagement shows up as a comparable loss of stimulus-locked variability reduction, independent of your existing hit/false-alarm-based Module C features.

STATISTICAL LEARNING & ODDBALL

Learning regularities in noise engages both neural predictive activity and representational changes

OPENALEX Coumarane Tirou et al. — Nature Communications — 2026-07-07

Corresponding: Romain Quentin — Université Claude Bernard Lyon 1 — France | [Open paper](#)

Using MEG during a visuomotor task with non-adjacent temporal dependencies embedded in noise, this study (with watch-listed statistical-learning researcher Dezső Németh as a co-author) dissociates two temporally separate learning mechanisms: a fast-emerging neural predictive-activity signal that appears before behavioral improvement, and a slower-building representational change that increases neural pattern similarity between the statistically dependent elements. This maps directly onto the bottom-up-statistical-learning-versus-top-down-template-matching framing at the core of the human chunk-in-stream task's design, and gives a concrete analysis template — pre-stimulus predictive decoding versus pattern-similarity growth across sessions — for separating those two processes in your own high-gamma and single-unit data as target sequences move toward full multi-template scene analysis.

Using hierarchical statistical learning models to model individual statistical learning

ARXIV Hanna Ringer et al. — arXiv (q-bio.NC) — 2026-07-07

Corresponding: Tatsuya Daikoku — University of Tokyo — Japan | [Open paper](#)

This preprint fits a Hierarchical Bayesian Statistical Learning model to individual EEG-derived learning trajectories as adults with and without dyslexia listen to structured tone sequences, and shows the model's simulated sequences closely match what participants actually generated — a working proof of concept for individual-level generative models of sequence learning. It's a useful methodological cousin to your own per-session model-fitting toolchain (the 24-model leaky-accumulator family fit by differential evolution, plus the tiered GLM strategy models): both try to recover an individual's latent learning process from sequence behavior, and this hierarchical Bayesian framing is worth comparing against your AICc/Akaike-weight model selection as an alternative way to handle per-mouse or per-session variability in the tone-sequence version of your task.

LANGUAGE, STROKE & BCI

Multi-network Topology Underlying Individual Language Learning Success

RSS:JOURNAL OF NEUROSCIENCE Journal of Neuroscience — 2026-07-08

Corresponding: Gangyi Feng — Chinese University of Hong Kong — Hong Kong SAR, China | [Open paper](#)

Using multimodal neuroimaging before and during seven days of artificial-language training across six tasks (auditory/speech categories, words, morphosyntax, sentence structure), this study finds that individual differences in learning outcome and rate are best predicted by the local efficiency of the dorsal-attention and frontoparietal control networks, not classic frontotemporal language regions alone. That distributed-network account of language-learning success is a relevant frame for the top-down template-matching side of the human chunk-in-stream task, particularly at the participant's frontoparietal recording sites (inferior and middle frontal gyrus): if attention/control-network engagement predicts how quickly a new stimulus class is learned, that argues for tracking frontoparietal signal quality across sessions when comparing performance across the harmonic-chord, vowel, and syllable stimulus classes.

Neural decoding of speech using deep neural ensembles

CROSSREF (MODEL-DISCOVERED) Seonghyun Yoon et al. — 2026-06-04

Corresponding: Francis R Willett — Stanford University — United States | [Open paper](#)

In a closed-loop test with a participant implanted with bilateral intracortical microelectrode arrays, this group shows that ensembling multiple independently trained decoders reduces speech brain-computer-interface word-error rate from 33.7% to 26.0%, and introduces a cheaper pseudo-ensembling method (test-time augmentation on a single decoder) that captures much of the benefit without the full computational cost. This is directly relevant to the human chunk-in-stream work's broader translational context — intracortical decoding in a person with severe language impairment — and the pseudo-ensembling trick is a concretely reusable idea for your own single-unit and high-gamma decoding pipelines in angular gyrus, supramarginal gyrus, and frontal sites, where training data per session is similarly limited by session count.

CIRCUITS, METHODS & POPULATION ANALYSIS

KIASORT: Knowledge-Integrated Automated Spike Sorting for Geometry-Free Neuron Tracking

RSS:JOURNAL OF NEUROSCIENCE Journal of Neuroscience — 2026-07-08

Corresponding: Kianoush Banaie Boroujeni — Princeton University — United States | [Open paper](#)

KIASORT is a new geometry-free spike-sorting algorithm that tracks each neuron's waveform drift independently rather than assuming a shared one-dimensional drift model across the probe, and the authors show it outperforms Kilosort4 on ground-truth benchmarks with heterogeneous per-neuron drift, validated on both primate and mouse data. Since your mouse Neuropixels 2.0 recordings from prelimbic and auditory cortex are currently sorted with Kilosort and drift is a known problem for long recordings, this is a direct pipeline-upgrade candidate worth benchmarking against your existing output — particularly if you've noticed unit-identity or waveform instability across a session, which per-neuron drift tracking is specifically designed to fix.

Neuropixels Opto: combining high-resolution electrophysiology and optogenetics

CROSSREF (MODEL-DISCOVERED) Anna A. Lakunina et al. — *Nature Methods* — 2026-06

Corresponding: Matteo Carandini — University College London — United Kingdom | [Open paper](#)

This paper introduces Neuropixels Opto, a probe integrating 960 recording sites with 28 independently addressable light emitters on the same shank, enabling spatially targeted optogenetic activation or silencing at specific cortical depths during high-density recording, and demonstrates efficient two-cell-type optotagging in mouse striatum. For the mouse sequence-discrimination task, where you already use Neuropixels 2.0 in prelimbic and auditory cortex, this would let you causally test which cortical layer or cell type carries the decision signal you currently only characterize correlationally — for instance, silencing superficial versus deep layers during the decision window to ask whether the prelimbic signal is more like the movement/evaluative activity described in today's medial-prefrontal-cortex threat paper, or a genuine causal driver of the go/no-go choice.

Quantifying Entrainment Evidence: A Comparison of Frequentist and Bayesian Approaches for Information Processing Pathway Maps

ARXIV Kaibo Zhang et al. — *arXiv (q-bio.NC)* — 2026-07-07

Corresponding: Andrew Thwaites — University College London — United Kingdom | [Open paper](#)

This preprint directly compares frequentist null-hypothesis testing against a Bayesian evidence-accumulation framework for adjudicating between competing computational models of an auditory processing pathway (loudness processing), arguing the Bayesian approach better handles collinear/competing models and yields a genuine measure of relative evidence rather than a simple reject/fail-to-reject decision. This is a close methodological parallel to your own model-selection practice — AICc and Akaike weights across your 24 nested drift-diffusion models, and cross-validated information gain for the GLM strategy models — and is worth reading as a check on whether a fully Bayesian evidence framework would handle collinearity between your structurally similar nested models (for instance leaky integration versus urgency) better than AICc currently does.

The structure of correlated variability reflects task-relevant information in sensory neurons.

PUBMED Ramanujan Srinath et al. — *PNAS* — 2026

Corresponding: Marlene R Cohen — University of Chicago — United States | [Open paper](#)

This group shows, via a circuit model plus multi-area population recordings, that the dominant axis of shared trial-to-trial variability in sensory neurons consistently aligns with task-relevant stimulus features and action plans, and that this shared-variability axis predicts the behavioral effect of microstimulation — reframing correlated variability as something that reveals rather than obscures the information sent downstream. This is a useful population-analysis framing for your own auditory cortex and prelimbic Neuropixels data: rather than treating trial-to-trial variability purely as noise to average out in dPCA or mixed-selectivity analyses, it's worth checking whether the low-dimensional axis of shared variability in your target-chord-evoked responses aligns with the decision-relevant dimension your GLM Module A features already isolate.

OTHER NOTABLE

Development of auditory and spontaneous movement responses to music over the first postnatal year.

PUBMED Trinh Nguyen et al. — *eLife* — 2026

Corresponding: Giacomo Novembre — Italian Institute of Technology — Italy | [Open paper](#)

This group simultaneously recorded EEG and markerless-pose-estimation-based body kinematics from 79 infants (3, 6, and 12 months) listening to music, finding that enhanced neural auditory responses to music are present from 3 months, but structured, complex movement patterns in response to music only emerge by 12 months — auditory encoding and motor response to sound develop on different timescales. This sits squarely inside your standing interest in continuous, pose-estimation-based behavioral quantification rather than per-trial outcomes, and offers a conceptual mirror for the mouse work: it's a reminder that a subject can show clear neural sensitivity to a sound sequence's structure well before that sensitivity is expressed in structured movement — relevant when interpreting mice that discriminate the target sequence from its reordered distractor in evoked activity but haven't yet generalized that to a reliable licking response, especially under the currently-running slow-pacing protocol.

PAPER OF THE DAY

Multiple event segmentation mechanisms in the human brain.

Tan T Nguyen et al. — eLife — 2026 — new pick — abstract only

Corresponding: Jeffrey M Zacks — Washington University in St. Louis — United States | [Open paper](#)

- Using fMRI and computational modeling, this group formally dissociates two distinct triggers for perceived event boundaries in continuous experience: high prediction error (mismatch between expected and observed information) and high prediction uncertainty (reduced confidence in the prediction itself) — accounts that prior work had largely conflated into one 'boundary' signal.
- The two boundary types produce distinguishable spatiotemporal signatures: error-driven boundaries evoke early pattern shifts in ventrolateral prefrontal areas followed by pattern stabilization in prefrontal and temporal areas, while uncertainty-driven boundaries evoke shifts confined to parietal regions of the dorsal attention network with minimal subsequent stabilization.
- Across both boundary types there is a conserved temporal cascade around human-identified boundaries — early pattern shifts in anterior temporal regions (about 11.9 s before the boundary), followed by shifts in parietal areas (about 4.5 s before), then whole-brain pattern stabilization (about 11.8 s after) — suggesting event segmentation is a distributed, sequenced process rather than a single mechanism firing at one moment.
- This gives chunking a genuine operational, two-factor definition (error versus uncertainty) with dissociable neural signatures and timing, rather than treating 'a boundary was detected' as a single undifferentiated event — directly useful for thinking about what a chunk boundary in an auditory stream actually is, computationally, not just perceptually.

HOW TO READ IT (≈20 min)

About 20 minutes, working from the abstract and title alone (the automated full-text fetch for this paper failed and returned unrelated content, so treat anything beyond the abstract as unverified until you pull the actual eLife PDF yourself — doi.org/10.7554/eLife.107955): start by re-reading the abstract slowly twice, since nearly every clause carries a distinct claim (two boundary-generating models, three region-by-timing dissociations, one shared cascade). When you get the real paper, prioritize the Results section describing the error-driven versus uncertainty-driven pattern-shift regions and their timing, and the Methods description of how the two computational models actually generate a 'boundary' from prediction error versus prediction uncertainty — that computational operationalization is the part most transferable to your own modeling toolchain. Skim the Introduction for how this reconciles with older event-segmentation-theory accounts, and skip any methods details specific to fMRI preprocessing/denoising unless you're troubleshooting your own imaging pipeline.

YOUR QUESTION TO ANSWER

In the human chunk-in-stream task, a Start-Target or End-Target trial gives you a natural candidate boundary — the transition into or out of the memorized template. If you tagged each such transition as more error-like (a template match violated by an unexpected element) or more uncertainty-like (a genuinely ambiguous, near-miss distractor) using this paper's operational definitions, would you predict your high-gamma responses at those transitions to show the same anterior-temporal-then-parietal cascade — and which of your five staged difficulty levels would give you the cleanest contrast to test whether that cascade's timing changes as the task gets harder?